

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');

```

```

fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off200GHminFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');

```

```

*****

```

```

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off200GHminFF

```

```

fprintf('*****\n');

```

```

*****

```

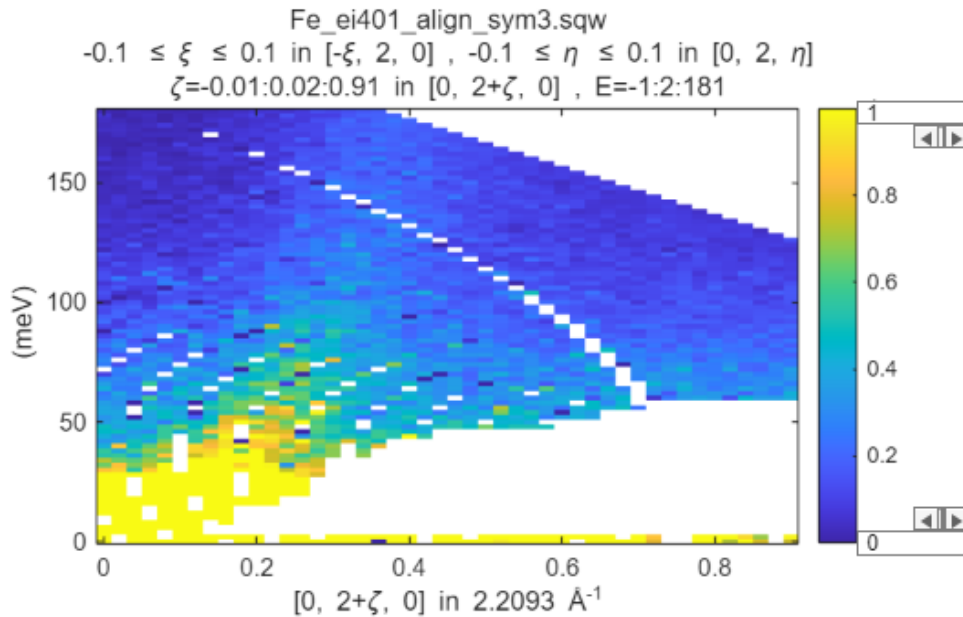
```

cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

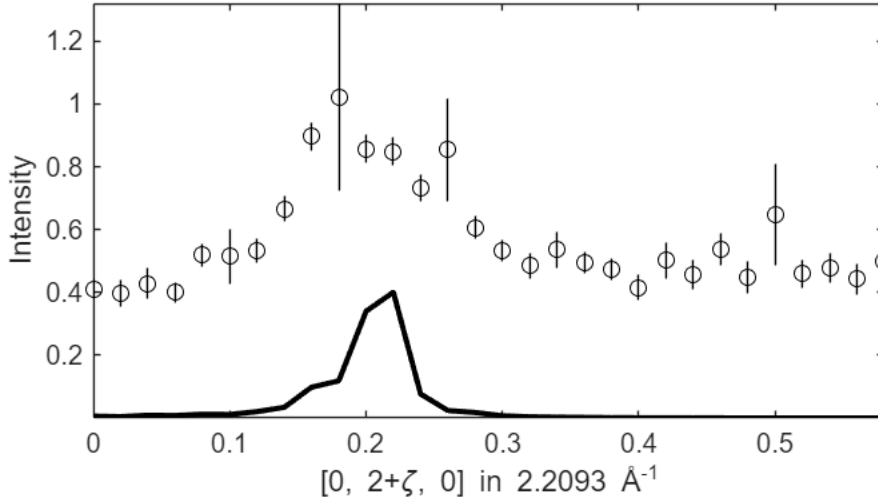
cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****
**** En = 45±5 going to 175
*****
```

Fe_ei401_align_sym3.sqw

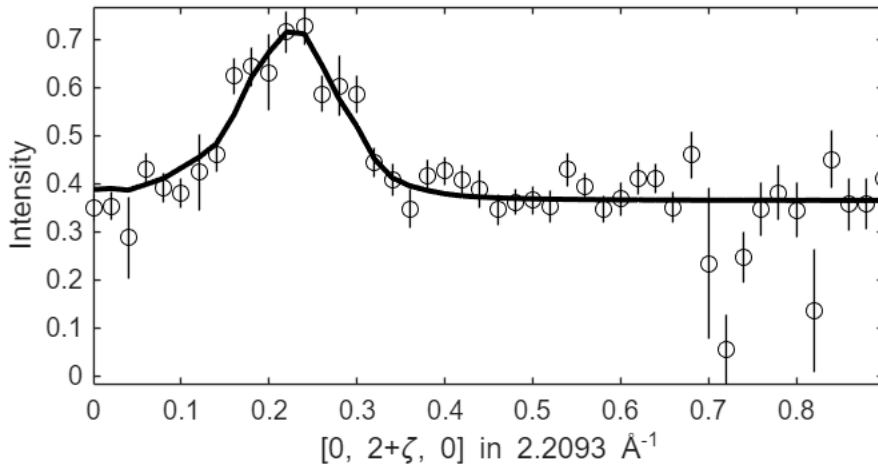
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $40 \leq E \leq 5$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



 **** En = 55±5 going to 175

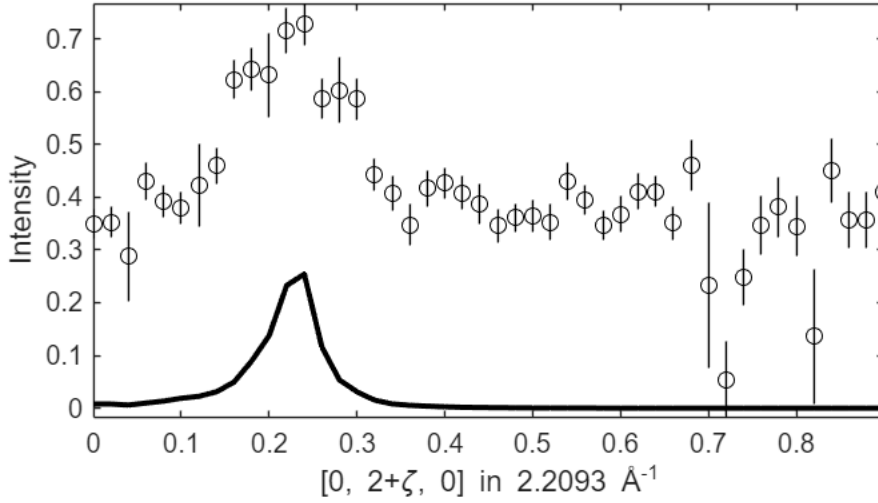
Fe_ei401_align_sym3.sqw

$S=0.73\pm0.13$; $J_0= 25\pm 1.1$; $J_1= 0\pm 0$; $\gamma=10.1571\pm5.07524$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $50 \leq E \leq 6$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



Fe_ei401_align_sym3.sqw

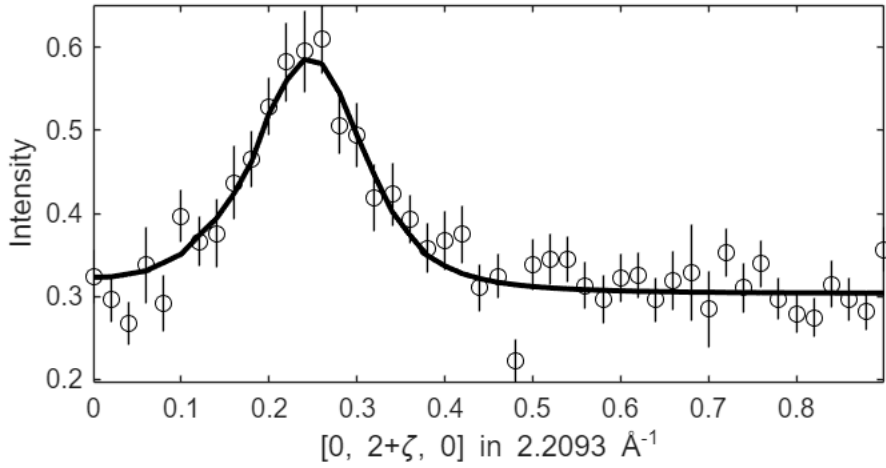
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $50 \leq E \leq 6$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



 **** En = 65 ± 5 going to 175

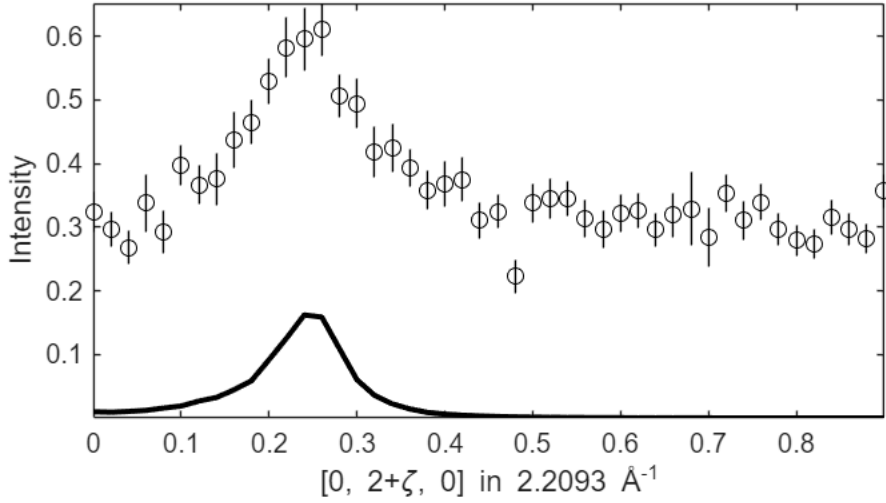
Fe_ei401_align_sym3.sqw

$S = 0.82 \pm 0.15$; $J_0 = 27 \pm 1.4$; $J_1 = 0 \pm 0$; $\gamma = 20.8476 \pm 7.30617$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $60 \leq E \leq 7$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



Fe_ei401_align_sym3.sqw

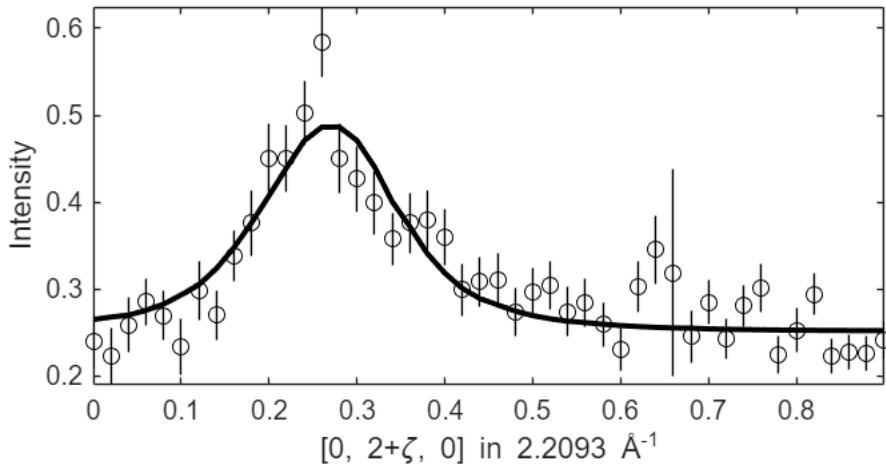
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $60 \leq E \leq 7$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



 **** En = 75 ± 5 going to 175

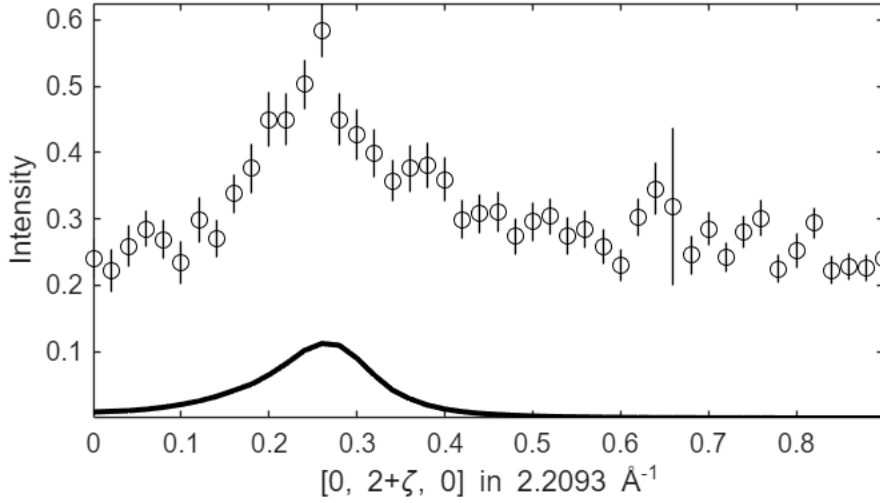
Fe_ei401_align_sym3.sqw

S= 1 ± 0.063 ; J0= 29 ± 1 ; J1= 0 ± 0 ; gamma= 38.2018 ± 2.49242
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $70 \leq E \leq 8$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



Fe_ei401_align_sym3.sqw

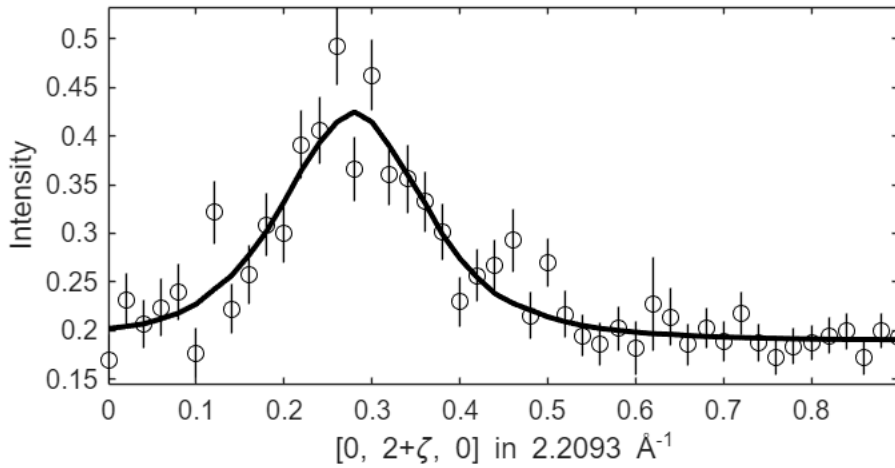
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $70 \leq E \leq 80$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



 **** En = 85±5 going to 175

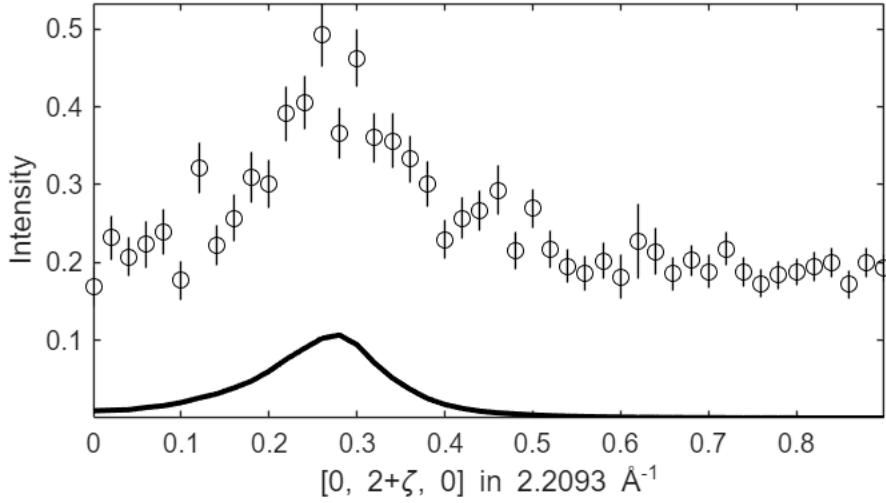
Fe_ei401_align_sym3.sqw

S= 1.2±0.29; J0= 31± 2.9; J1= 0± 0; gamma=48.2575±17.697
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $80 \leq E \leq 90$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



Fe_ei401_align_sym3.sqw

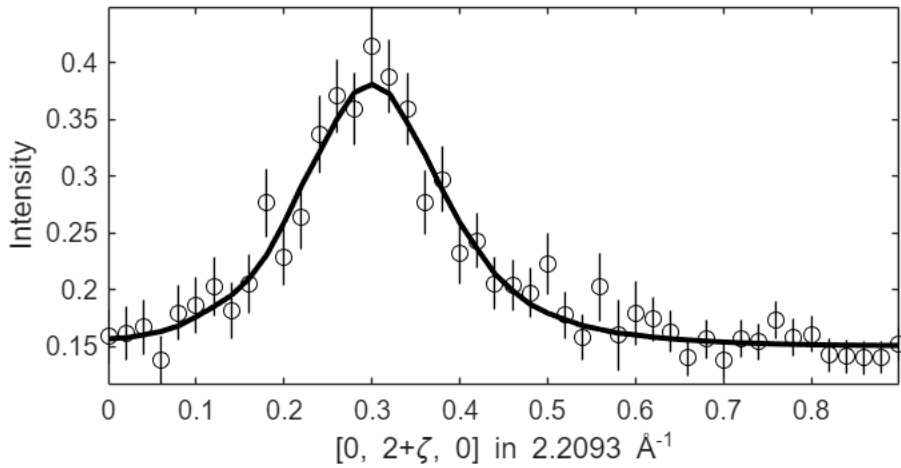
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $80 \leq E \leq 90$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



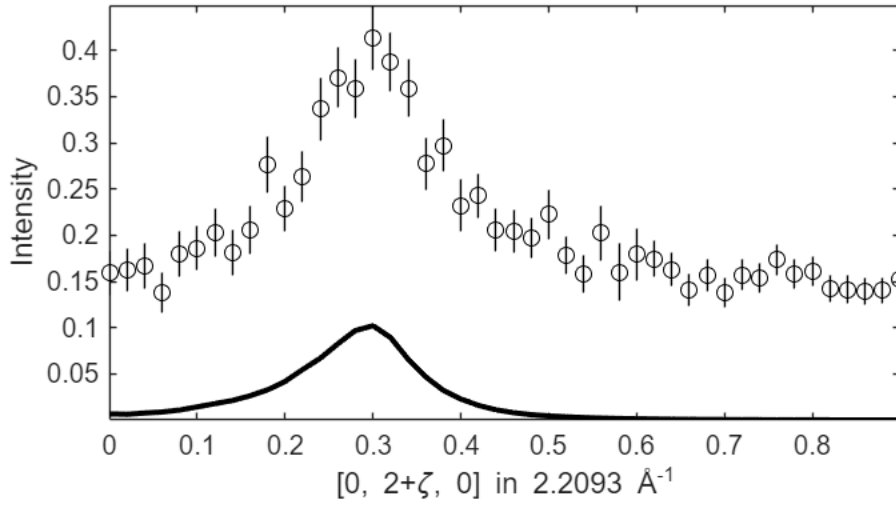
 **** En = 95±5 going to 175

Fe_ei401_align_sym3.sqw

S= 1.1±0.036; J0= 30±0.54; J1= 0± 0; gamma=43.0384±1.37242
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $90 \leq E \leq 100$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

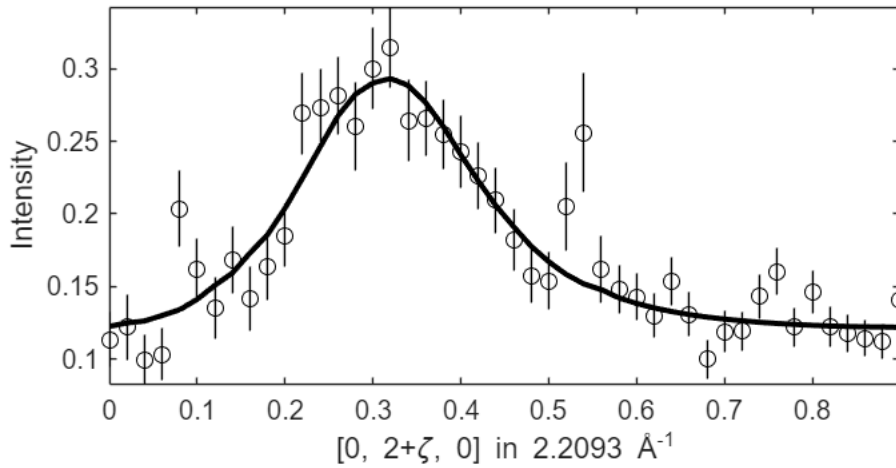


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $90 \leq E \leq 110$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

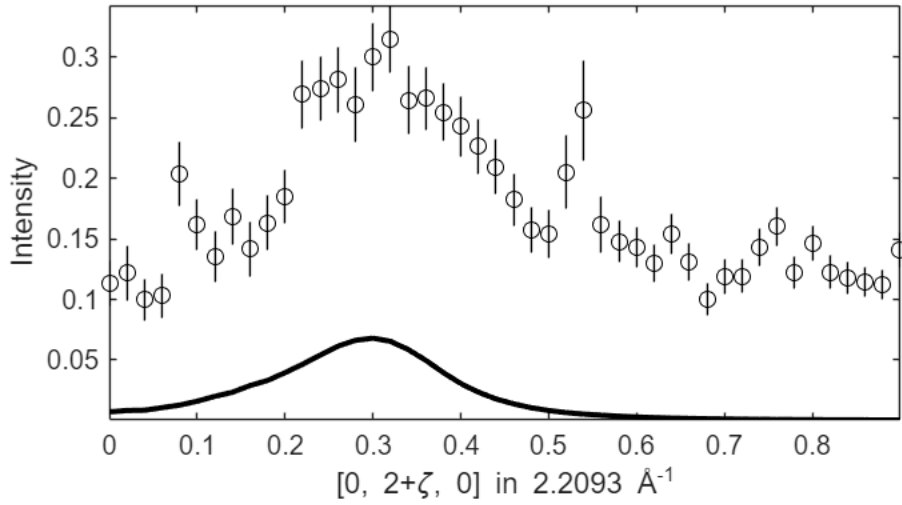


 **** En = 105±5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.5 \pm 0.65$; $J_0 = 37 \pm 7.6$; $J_1 = 0 \pm 0$; $\gamma = 100.673 \pm 58.7557$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $100 \leq E \leq 110$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

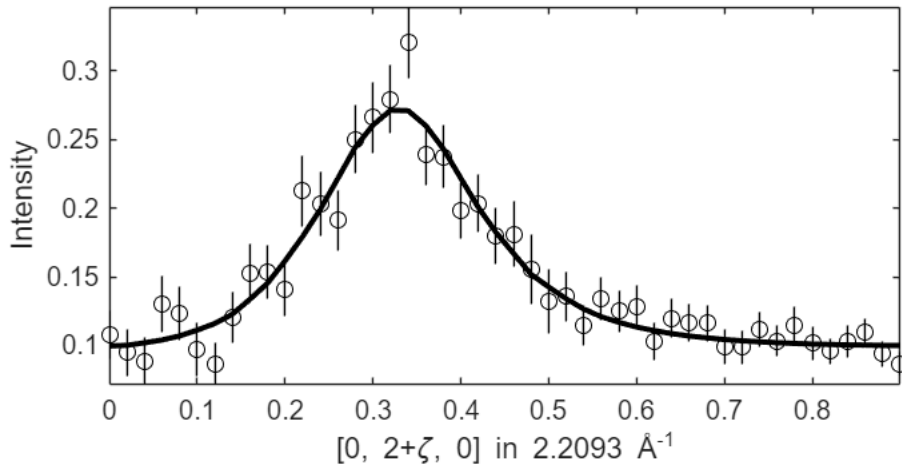


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $100 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

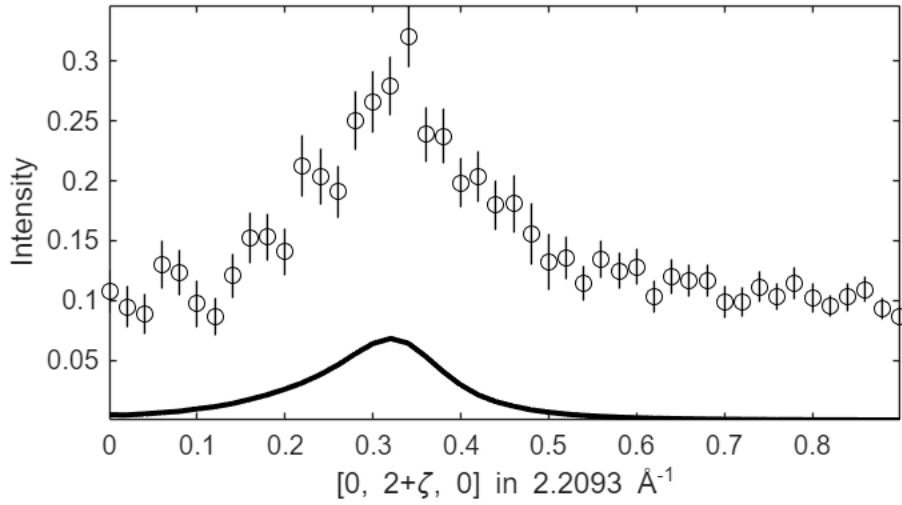


 **** En = 115±5 going to 175

Fe_ei401_align_sym3.sqw
 S= 1 ± 0.038 ; J0= 32 ± 0.67 ; J1= 0 ± 0 ; gamma=59.2682±1.98411
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $110 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

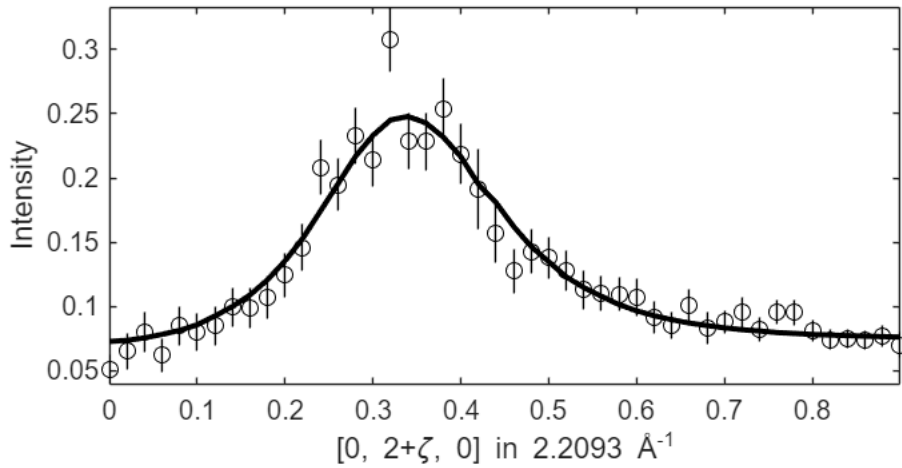


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $110 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

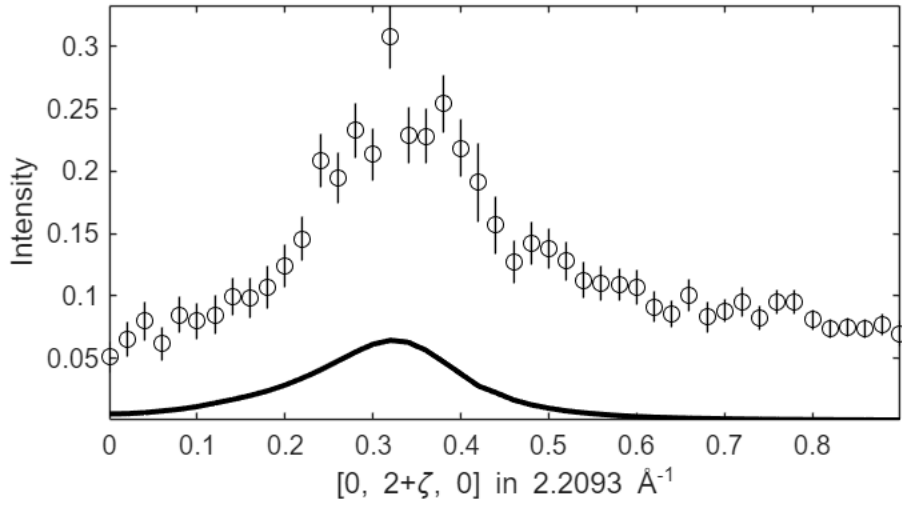


 **** En = 125±5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.4 \pm 0.27$; $J_0 = 35 \pm 2.9$; $J_1 = 0 \pm 0$; $\gamma = 86.4686 \pm 23.6464$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $120 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

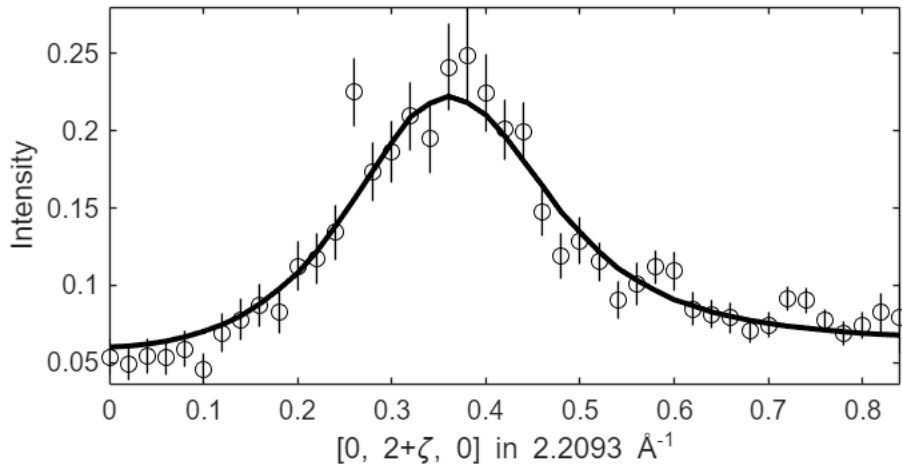


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $120 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

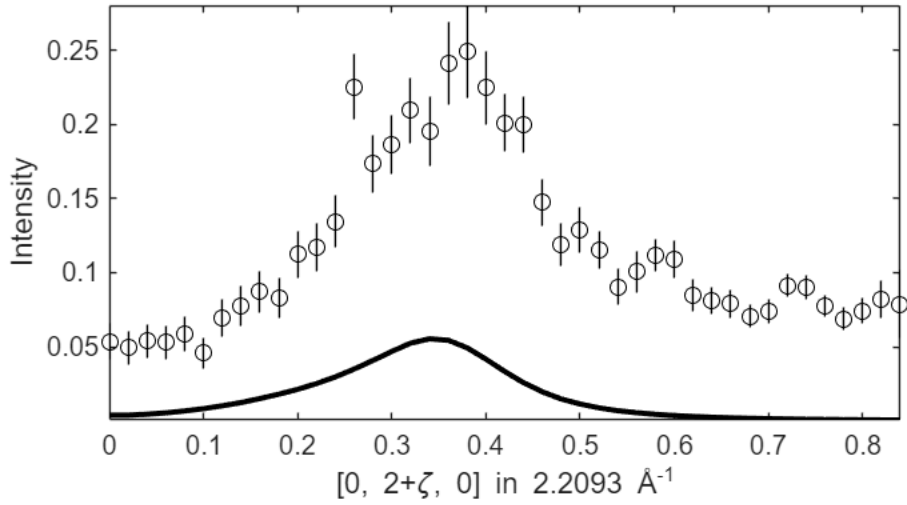


 **** En = 135 ± 5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.28$; $J_0 = 33 \pm 3.1$; $J_1 = 0 \pm 0$; $\gamma = 82.6731 \pm 26.6523$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $130 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

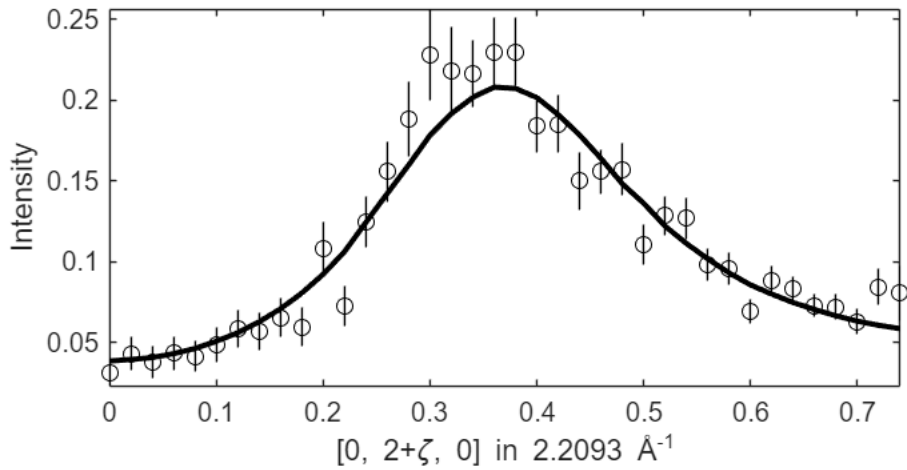


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $130 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

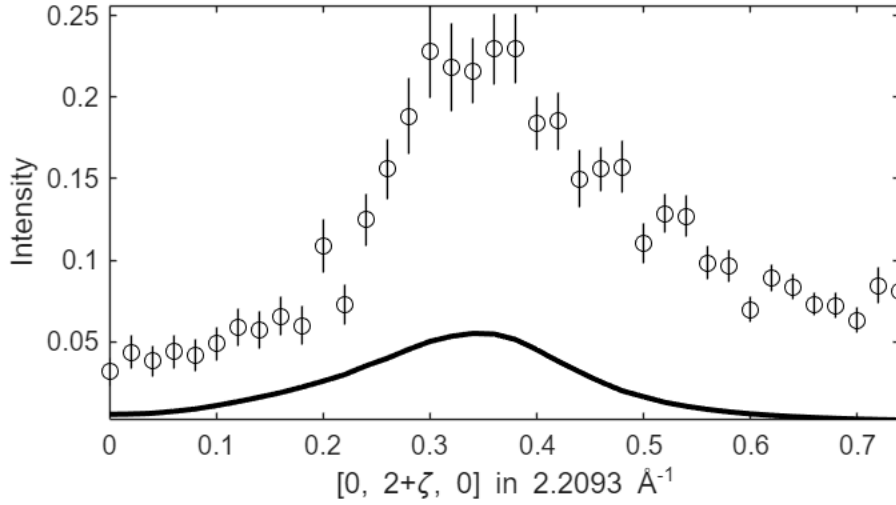


 **** En = 145 ± 5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.8 \pm 0.14$; $J_0 = 39 \pm 0.75$; $J_1 = 0 \pm 0$; $\gamma = 132.034 \pm 8.32506$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $140 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

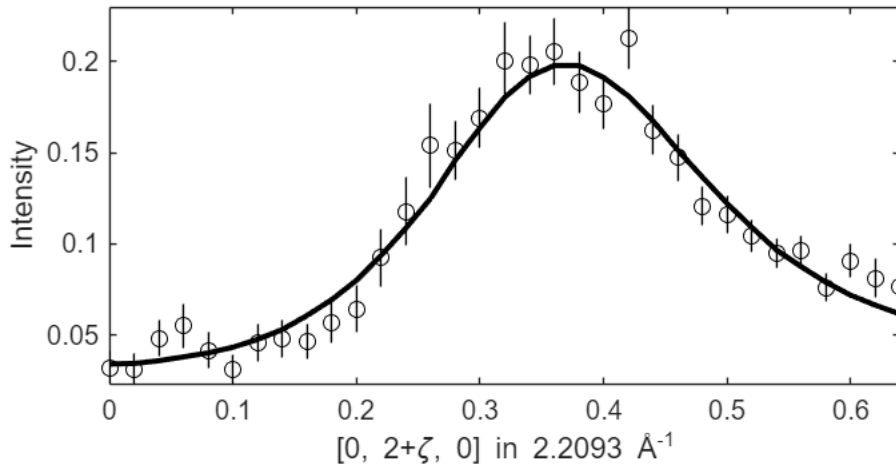


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $140 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

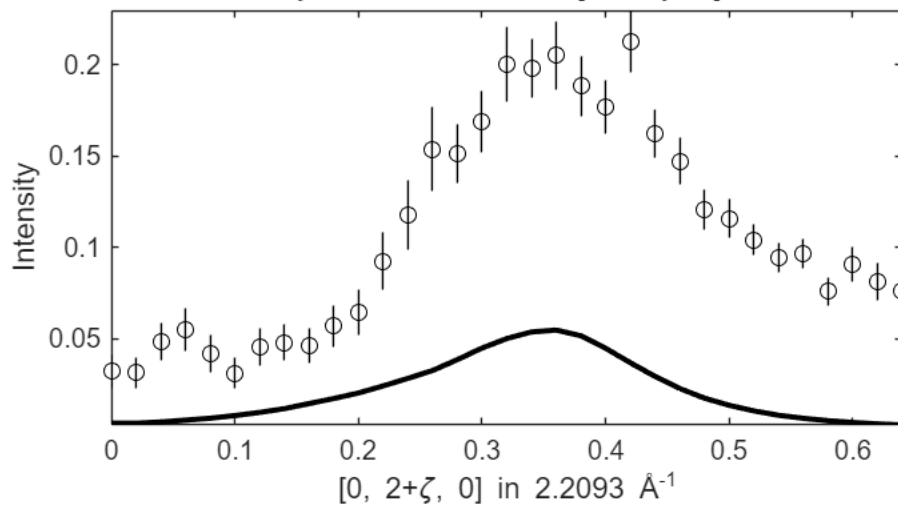


 **** En = 155±5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.4 \pm 0.26$; $J_0 = 37 \pm 2.7$; $J_1 = 0 \pm 0$; $\gamma = 97.5402 \pm 23.0314$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $150 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

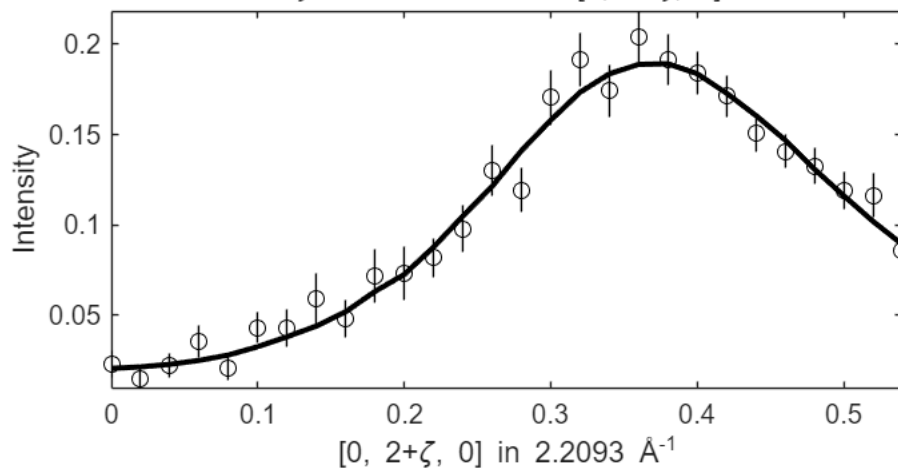


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $150 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

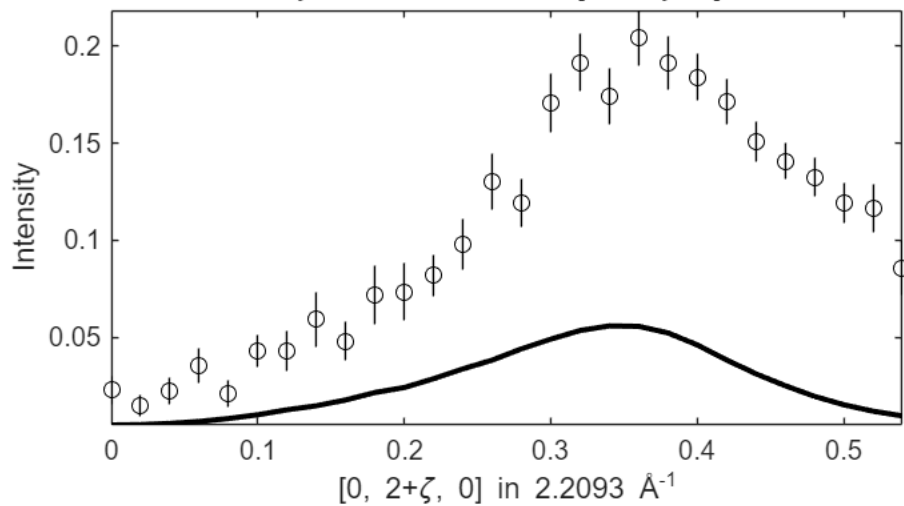


 **** En = 165±5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.9 \pm 0.31$; $J_0 = 42 \pm 3.1$; $J_1 = 0 \pm 0$; $\gamma = 134.41 \pm 29.7322$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $160 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

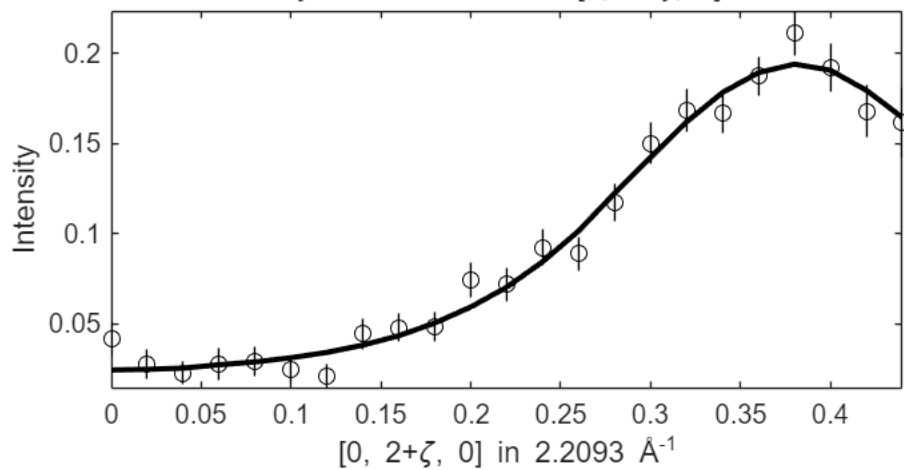


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $160 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$

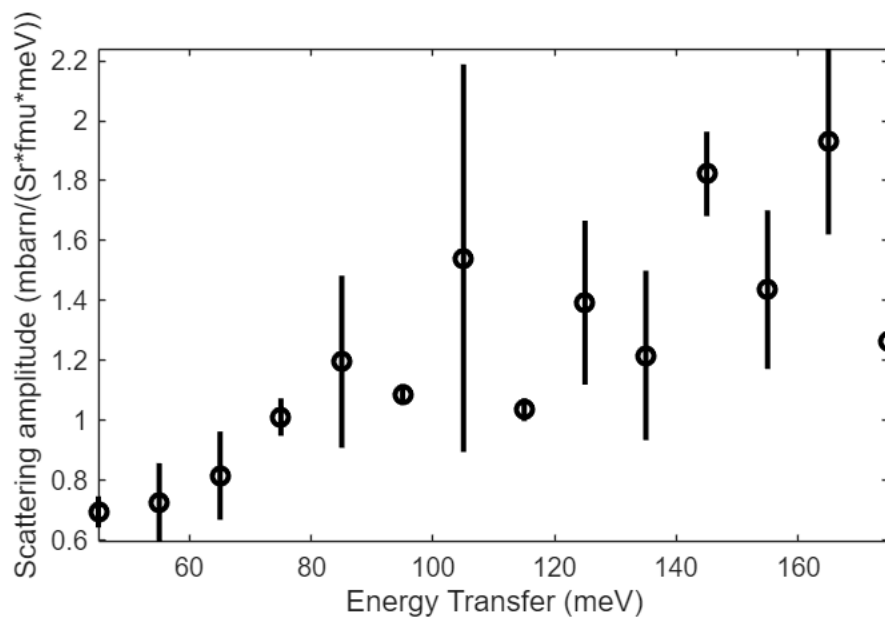
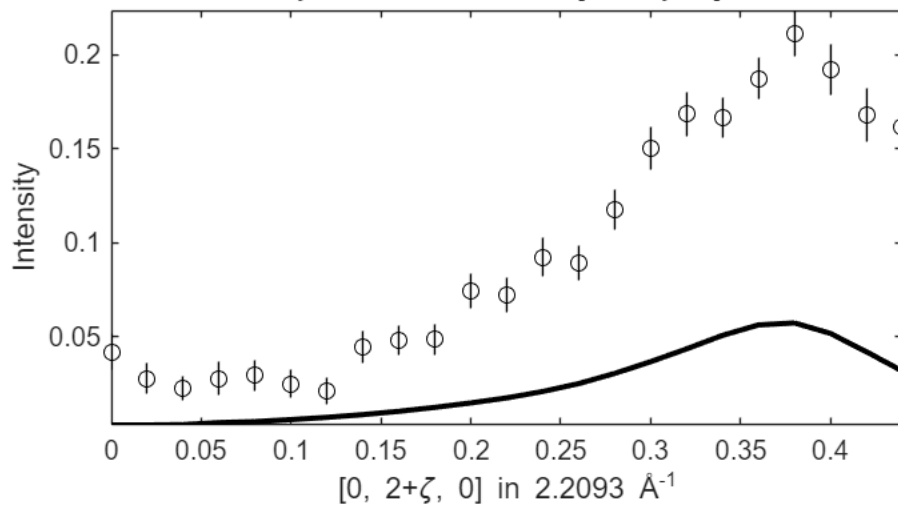


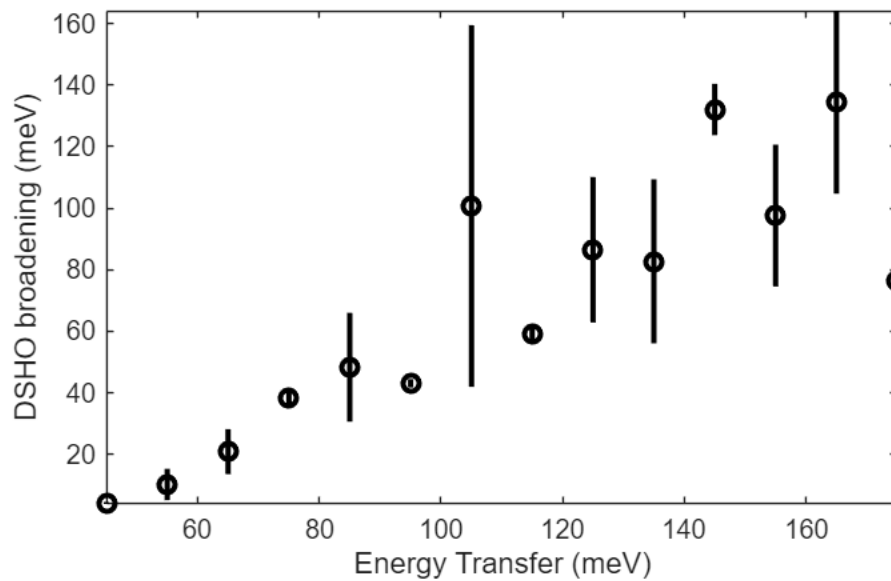
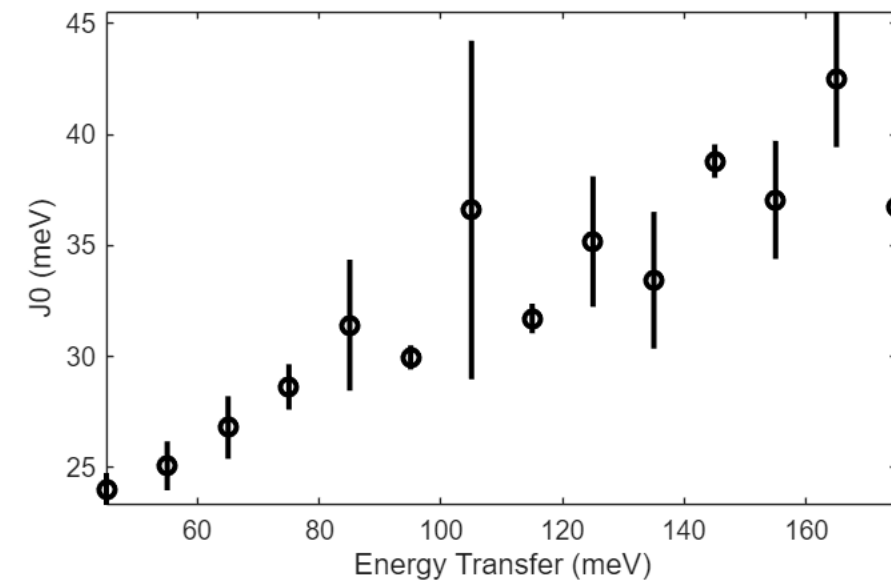
 **** En = 175±5 going to 175

Fe_ei401_align_sym3.sqw
 $S = 1.3 \pm 0.099$; $J_0 = 37 \pm 0.98$; $J_1 = 0 \pm 0$; $\gamma = 76.4275 \pm 6.70411$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $170 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$



Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $170 \leq E \leq 1$
 $\zeta = -0.01:0.02:0.91$ in $[0, 2+\zeta, 0]$





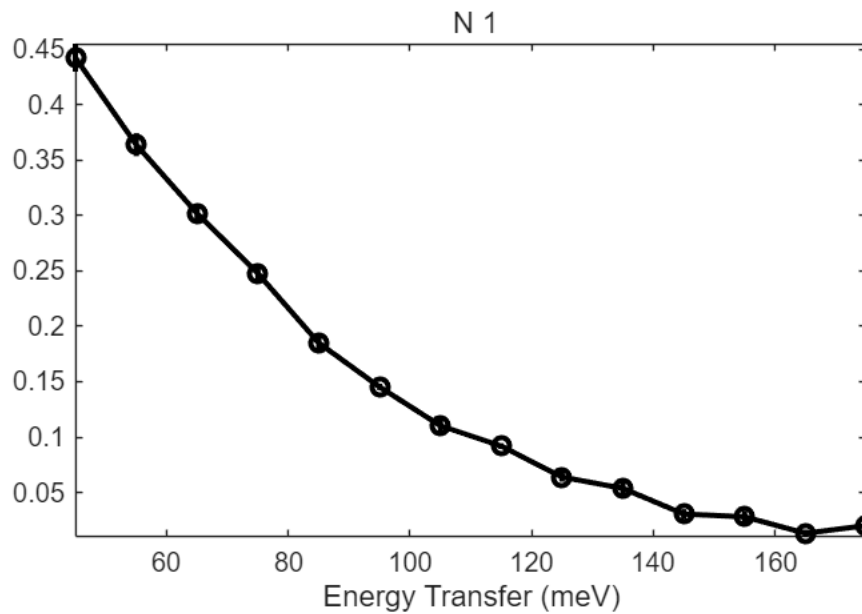
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

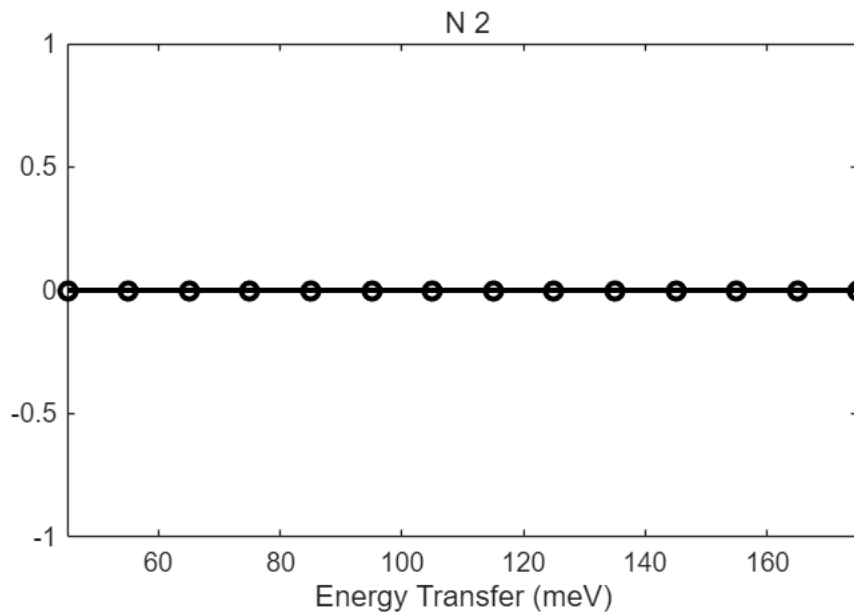
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;  
save(CutFitFileGH,"fit_src_struct",-v7.3');  
close all;  
%end
```